GIACOMO LOMBARDO

Computer Engineer | AI and Deep Learning Focus

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Professional Profile

Computer Engineer pursuing a Master's Degree (expected graduation July 2026), specializing in **Artificial Intelligence**. Driven by a strong commitment to continuous learning, problem-solving, and analytical thinking, I have extensive experience in teamwork, having completed most of my academic projects in groups of up to five people. Throughout my academic career, I have developed a deep understanding of **Machine Learning** and **AI** techniques, focusing primarily on **Computer Vision** and **NLP**. I am proficient in frameworks such as PyTorch and TensorFlow, a skill set further refined through various academic projects and my Master's thesis, which focuses on the optimization of Vision Transformers using Pruning and Neural Architecture Search (NAS). I am highly motivated to contribute to the development of innovative AI/ML solutions within a stimulating professional environment.

Technical Skills

- **Artificial Intelligence:** Computer Vision, Natural Language Processing (NLP), Deep Learning, Machine Learning.
- **AI Frameworks and Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy.
- **Programming Languages:** Python (Focus on AI & Data Science), Java, C, JavaScript.
- **Databases, Data, and Systems:** Graph Databases (Neo4j), Relational Databases (SQL), JSON, XML, Linux, Windows.
- **Tools and IDEs:** PyCharm, Visual Studio Code, Git.

Relevant Projects

- **ViT Compression Framework** | *PyTorch*
  Design and development of a framework for the compression of Vision Transformer models. Optimization of computational efficiency through Neural Architecture Search (NAS) and pruning techniques.
- **DeepFake Detection with Attention** | *TensorFlow, Keras*
  Development of a system for deepfake image recognition. Initially implemented using classical ML models, then through standard CNN architectures and CNNs equipped with **Attention** mechanisms.

- **SVD CNN Optimization** | *MATLAB*
  Mathematical analysis and implementation of Singular Value Decomposition (SVD) to evaluate the effects of dataset compression on the classification performance of a Convolutional Neural Network.
- **Face Morphing** | *Python, OpenCV*
  Computer Vision system for the automatic generation of smooth video transitions (morphing) between faces starting from two static input images.
- **Software Engineering and Full-Stack Development** | *Java, Spring Boot, Kotlin, SQL*
  Development of several end-to-end software applications, including a sports management system with UML documentation (Java/MySQL), a workshop management platform with a Spring Boot back-end and jQuery front-end, and a native Android app in Kotlin.

Professional Experience

AI Developer / Intern — **July 2023 – October 2023**
ITALTEL S.p.A. — *Carini (PA)*
Curricular Internship.

- **Main Project:** Design and development of Computer Vision software dedicated to the automatic recognition and counting of labels in photographic inputs.
- **Stack and Skills:** Development and training of Machine Learning models using **TensorFlow** and **Keras** APIs.
- **Architectures:** Design, implementation, and training of **Convolutional Neural Networks (CNN)** for image analysis and classification.

Education and Training

Master's Degree in Computer Engineering (LM-32) — **2024 – Ongoing**
University of Palermo — *Palermo*
- Thesis based on Transformer model compression through pruning and architectural search techniques.

Bachelor's Degree in Computer Engineering (L-8) — **2021 – July 2024**
University of Palermo — *Palermo*
- Final grade: **110/110 cum laude**

High School Diploma (Applied Sciences) — **2016 – 2021**
Michele Cipolla Scientific High School — *Castelvetrano (TP)*
- Final grade: **100/100 cum laude**

Certifications and Languages

- **English:** B2 Level.
- **Italian:** Native Speaker.

I hereby authorize the processing of my personal data in accordance with Legislative Decree June 30, 2003, n. 196 and the GDPR (EU Regulation 2016/679).